

b. Compute ACF and PACF.

16 2 3 2

t	1	2	3	4	5	6	7	8	9	10
x _t	71	72	69	68	64	65	72	78	75	75

14.a.i. Discuss the stationarity, invertibility and causality of $x_t = -0.2x_{t-1} + 0.15x_{t-2} + \varepsilon_t + \varepsilon_{t-1} + 0.25\varepsilon_{t-2}$.

16 2 4 2

ii. Describe the statistical properties of AR, MA, ARMA.

(OR)

b. Compute AIC value for

16 2 4 2

t	0	1	2	3	4	5	6	7	8	9
x _t	4	5	6	7	9	11	4	6	9	3

By assuming ARMA (2,1) with $\phi_1=0.2$, $\phi_2=1.2$, $\theta_1=0.1$, $\sigma^2=1.5$.

15.a. Compute $\gamma_0, \gamma_1, \gamma_2$ and ρ_0, ρ_1 for $x_t = \phi_1 z_{t-1} + \phi_2 z_{t-2} + \varepsilon_t - \theta_1 \varepsilon_{t-1}$ with

16 2 5 2

$$\phi_1 = \begin{bmatrix} -0.816 & 0.116 \\ 0.1 & -0.623 \end{bmatrix}, \theta_1 = \begin{bmatrix} -0.801 & 0 \\ 0 & -1.623 \end{bmatrix}, \phi_2 = \begin{bmatrix} -0.643 & 0.615 \\ 0 & -0.592 \end{bmatrix}$$

and

$$\Sigma_\varepsilon = \begin{bmatrix} 4 & 2 \\ 2 & 5 \end{bmatrix}$$

(OR)

b. Describe briefly about VAR model.

16 2 5 2

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